

LAB 3 - ARCHITECTURE SELECTION & COMPONENT MODEL

Smart Lab Equipment & Slot Reservation Portal
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1. Architecture Selection

We chose Layered Architecture for the Smart Lab Equipment & Slot Reservation Portal. The system separates presentation, business rules, and data management into clear layers, matching the portal workflow documented in Labs 1 and 2.

2. Architectural Style Analysis

Style	Relevant advantages	Relevant limitations
Layered	Clear separation; easy to enforce reservation/calibration rules; suitable for a transactional portal.	Some inter-layer overhead; scaling is less granular.
Microservices	Independent deployment/scaling and fault isolation.	Adds operational/network complexity and distributed-data consistency concerns for a small controlled portal.
Client-Server	Centralized control and straightforward data consistency.	Less separation of responsibilities; the centralized server can become a bottleneck.

3. Two Scenario-Specific Reasons

1) The core rules are transaction-heavy and centralized: a reservation must be at most 2 hours, within 7 days, use calibrated equipment, and reject an occupied slot. Keeping these rules in one business layer makes them consistent and testable. 2) The requirements emphasize immediate calibration visibility, cancellation, returns, and concurrent reservation control, which fit a single application with a shared data layer and strict database locking.

4. Security Advantage

Authentication and role authorization are isolated in the business layer. Every reservation or calibration-management request can be checked before the operation reaches the data layer, directly supporting the requirement that unauthorized users must be denied access.

5. Performance Benefit

The architecture keeps the UI responsive while the business layer performs focused validation and the data layer handles indexed, transactional reads/writes. Database-level locking can protect the reservation transaction from races while supporting the stated 200 ms concurrent-request target.

6. Component Model

The diagram contains six components: Student/Technician UI, Authentication & Authorization, Reservation Manager, Equipment & Calibration Manager, Return & Late-Return Manager, and Lab Reservation Database. Interfaces include IAuthAPI, IReservationAPI, IEquipmentAPI, IReturnAPI, and IRepository, with provided/required UML notation and labeled dependencies.

Source basis: Lab 1 requirements and use-case flow; Lab 2 UML/use-case artifact; Lab 3 architecture/component-modelling handout.